SIMVASTATIN

Take in the evening

Many CYP3A4 interactions

CHEMICAL NAME (1S,3R,7S,8S,8aR)-8-{2-[(2R,4R)-4-hydroxy-6-oxooxan-2-yl]ethyl}-3,7-dimethyl-1,2,3,7,8,8a-hexahydronaphthalen-1-yl 2,2-dimethylbutanoate

CLASS HMG-CoA reductase inhibitor (statin), moderate intensity

TRADE NAMES Zocor · Simvador · Inegy (with ezetimibe)

THERAPEUTIC USES / INDICATIONS

- Hypercholesterolaemia.
- Prevention of cardiovascular events (MI, stroke).
- Familial hypercholesterolaemia (heterozygous).

ADMINISTRATION

- **PO** once daily in the evening (short half-life; cholesterol synthesis peaks at night).
- Prodrug (lactone).

DOSAGES

- **Adult:** 20–40 mg once daily at night.
- 80 mg not recommended for new patients (myopathy).
- With amlodipine, diltiazem: max 20 mg; with verapamil: max 20 mg (some guidance 10 mg); with amiodarone: max 20 mg.
- Contraindicated with clarithromycin, itraconazole, ketoconazole, HIV PIs.

PHARMACOKINETICS

- **Onset:** LDL falls in 2 weeks. Bioavailability <5% (extensive first pass).
- Lactone prodrug hydrolysed to active acid; metabolised by CYP3A4.
- Excreted in bile. **t½:** 2–3 h.

PHARMACODYNAMICS

- Inhibits **HMG-CoA reductase** → ↓ cholesterol synthesis, ↑ LDL receptors → ↓ LDL-C (30–40%).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Myalgia, GI upset.
- Headache.
- Raised transaminases.

ADVERSE EFFECTS (SERIOUS)

- Myopathy and rhabdomyolysis (highest risk among common statins at 80 mg and with interacting drugs).
- Hepatotoxicity (rare).
- Interstitial lung disease (rare).

CONTRAINDICATIONS & CAUTIONS

- Active liver disease.
- Pregnancy, breastfeeding.
- Strong CYP3A4 inhibitors (clarithromycin, erythromycin, itraconazole, ketoconazole, HIV PIs), gemfibrozil, ciclosporin, danazol.

KEY INTERACTIONS

- **Clarithromycin, itraconazole:** contraindicated; withhold simvastatin during course.
- **Amlodipine, diltiazem, verapamil, amiodarone:** dose caps.
- **Grapefruit juice:** avoid.
- **Fusidic acid:** fatal rhabdomyolysis.
- **Warfarin:** ↑ INR.

NURSING MANAGEMENT

- Give in the evening.
- Review all interacting medicines on admission.
- Hold during clarithromycin courses if instructed.
- Monitor lipids, LFTs; ask about muscle pain.

PATIENT EDUCATION

- Take at night.
- Avoid grapefruit juice.
- Report muscle pain or dark urine.
- Tell doctors/pharmacists you take simvastatin before new antibiotics.

KEY POINTS

- Moderate-intensity statin; evening dose.
- Never with clarithromycin or itraconazole.
- CYP3A4 — many interactions; dose caps.
- Avoid 80 mg.